Preliminary Report
And Design Document

Team: PROMETHEUS

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Contents

1 INTRODUCTION **2**
1.1 Purpose of The mission . . . 2
1.2 TEAM ORGANISATION AND ROLES . . . 2
1.3 MISSION OBJECTIVES . . . 2

2 CANSAT DESCRIPTION **3**
2.1 MISSION OVERVIEW . . . 3
2.2 MECHANICAL/STRUCTURAL DESIGN . . . 3
2.3 ELECTRICAL DESIGN . . . 7
2.4 POWER BUDGET . . . 8
2.5 SOFTWARE DESIGN . . . 8
2.6 RECOVERY SYSTEM . . . 8
2.7 GROUND SUPPORT EQUIPMENT . . . 9

3 PROJECT PLANNING **9**
3.1 Time schedule of the CanSat preparation . . . 9
3.2 RESOURCE ESTIMATION . . . 9
3.2.1 BUDGET . . . 9
3.2.2 EXTERNAL SUPPORT . . . 10
3.3 TEST PLAN . . . 10
3.4 TIME MANAGEMENT . . . 10
3.5 RISK MANAGEMENT . . . 11

4 DATA ANALYSIS AND OUTREACH **11**
4.1 DATA ANALYSIS PLAN . . . 11
4.2 OUTREACH PROGRAM . . . 12

5 CONCLUSION **12**
5.1 SUMMARY OF THE PDR . . . 12
5.2 RECOMMENDATIONS FOR NEXT STEPS . . . 13

1 INTRODUCTION

1.1. Purpose of The mission

The principal goal of the CanSat mission is to collect environmental data through a series of probes in order to properly assess the status of the airspace the CanSat travels through. Potential impact ranges from environmental studies to pollution statistics with possible use cases such as air traffic and control and meteorological updates for civilian and military applications.

1.2. TEAM ORGANISATION AND ROLES

- Team Name: Prometheus. We chose our team name after the titan god of fire, who gave the secret of fire to mankind according to Greek mythology. A fitting name perhaps, for a development team regarding rocketry in Romania, given the lack of development accorded to this sector in our country.
- Team Composition: Our team is composed of Toma Lucas-Dan (Team Lead), Dinu Teodor Gabriel, Atodiresei Tudor-Ovidiu, Croitoriu Alexandru, and Pintilie Dragos.
- Background: Our background is one of common nature, as we are all representatives of the Babes-Bolyoi Physics and the UTCN ETTI departments respectively. All of us have, at one point, expressed an interest and experimented with rocket technology, and have developed a positive attitude towards working in the aforementioned domain.
- Field of work: Our field of work is relatively mixed, as due to our similar skill sets, workloads are easily interchanged. A versatile skill set has allowed us to interchange tasks based on demand or difficulty, as oftentimes during the development of our project, we have switched design priorities in order to overcome technical hurdles. That being said however, it is fair to mention that Croitoriu Alexandru and Pintilie Dragos are typically in charge of 3d modelling, Atodiresei Tudor-Ovidiu assisting with animations and technical overview, and Toma Lucas-Dan with Autodiresei Tudor-Ovidu completing the electronics integration and software programming side of the project.
- Expected workload: Our team has the general expectation for each member to accomplish their given tasks, assigned to the team lead's best ability in a fair and rational manner. Oftentimes we have encountered a team member that has accomplished their task ahead of schedule - and due to our fluid team composition and tight-knit teamwork, we have managed to transition focus to remaining projects seamlessly.
- Hours dedicated: We estimate the current dedicated work time at 120 Hours split between 5 team members as of the moment of writing.
- Team's Activity: During the development of our project so far, our team has met on a bi-weekly basis in order to discuss and further project potential design ideas for the CanSat design and development.

1.3. MISSION OBJECTIVES

1. Our choice of secondary mission was an unusual one. Instead of attempting to fit more sensors into a slow descent vehicle, our team opted for a more unconventional approach through the use of three (3) slow descent sub-probes, launched from the main body of the principle CanSat with the minimum requirements to fulfill the principle objectives: air pressure, pollution, and temperature. The projected estimate is that we will be able to create a three dimensional graph of air vectors, pressure differences, temperature differences and pollution

flow through the use of multiple probes, thereby collecting vectorial data respective to the required atmospheric data.
2. Successful CanSat launch and recovery would be characterized through the following criteria: successful launch, sub-probe deployment, successful real-time data transmission, and recovery of all components related to the CanSat project.
3. Our research expectations would be best represented by the varied sub-probes and their properties, as their expected performance may vary from their real-life performance, as well as the ability to graph pressure differences and temperatures as vectors, along with predicting pollution vectors. We hope to better our system through testing, as the ability to predict the aforementioned vectors in a package as small as the CanSat would prove useful in many domains.
4. Our expected tests are to measure the efficiency with which a dispersed slow-descent probe can project and predict pollution changes and pressure differences over a small area, as well as the ability to deploy and disperse said sub-probes over a wider area with positive results. We intend to contrast our direct results with meteorological data published on behalf of the European Union's Meteorological Infrastructure System, or EMI, through the use of EUMETSAT, a coordinated effort among countries in the EU for meteorological survey and use.

2 CANSAT DESCRIPTION

2.1. MISSION OVERVIEW

The mission will be carried out by designing and building a CanSat that will be launched and deployed from a rocket at an altitude of about 1200 metres. The CanSat is projected to descend no faster than 12 metres per second. The CanSat will separate from the launch rocket at approximately 1150 meters, after which it will subsequently deploy a set of three (3) sub-probes that will graph air pressure, temperature, and pollution vectors in real time, and send the aforementioned data back to a receiver on the ground. Lower altitude deployments are not expected to pose any real risk, as on a theoretical level, full deployment and arrested descent of the three sub-probes are expected to complete within 15 to 20 meters of initial canister deployment. Key elements that will be used to accomplish the mission objectives include:

- Sensors for receiving data on the following: air pressure, temperature, and gas composition (CO_2 and CH_4)
- Processing unit for aforementioned sensors
- Deployment mechanisms for deploying the three (3) sub-probes
- Transmission elements contained within the three (3) sub-probes for real-time telemetry

2.2. MECHANICAL/STRUCTURAL DESIGN

1. Mechanical Design: The CanSat structure is made of lightweight biodegradable PLA plastic printed in parts through the use of a 3D printer, which provides strength and durability while keeping the overall weight of the CanSat at a minimum, as well as providing a biodegradable structure in the unfortunate event that a sub probe is lost. The structure of the three (3) individual sub-probes is designed to be completely contained at launch, insuring a water-tight and resistant quality to the components residing within that guarantees safe storage and flight integrity, minimizing any present risks of deployment failure.

2. Components: The major components of the CanSat include the main board, sensors, transmitter, and battery. The main board houses the microcontroller and provides an interface between the sensors and the transmitter, the sensors including a temperature sensor, a pressure sensor, and a pollution sensor. The transmitter is used to send data back to the ground station, while the battery is used to power the entirety of the circuit for the duration of the flight.
3. Placement: The main board is housed directly adjacent to the left of the spring designed for full deployment of the CanSat sub-probes, the battery and sensor array being mounted around the main board, while the communications and pollution sensors will be mounted below and above the main board respectively
4. Technical and Design Schematics

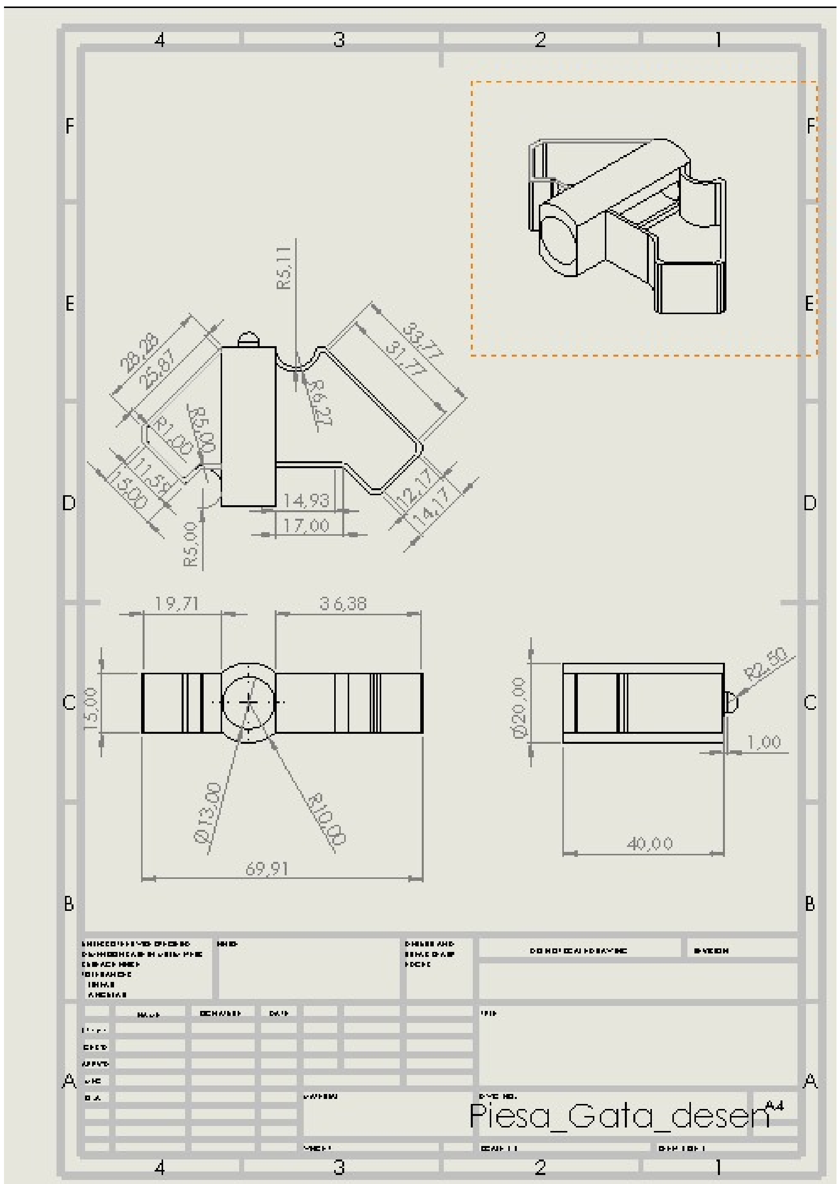

Figure 1: 1.01

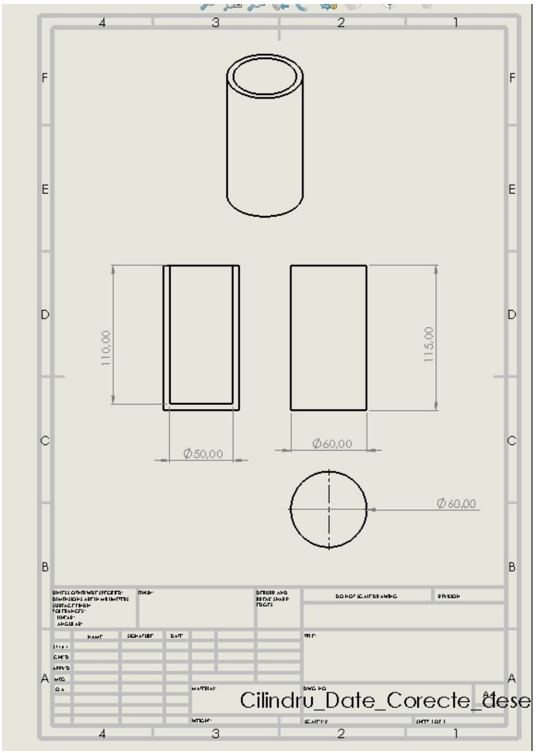

Figure 2: 1.02

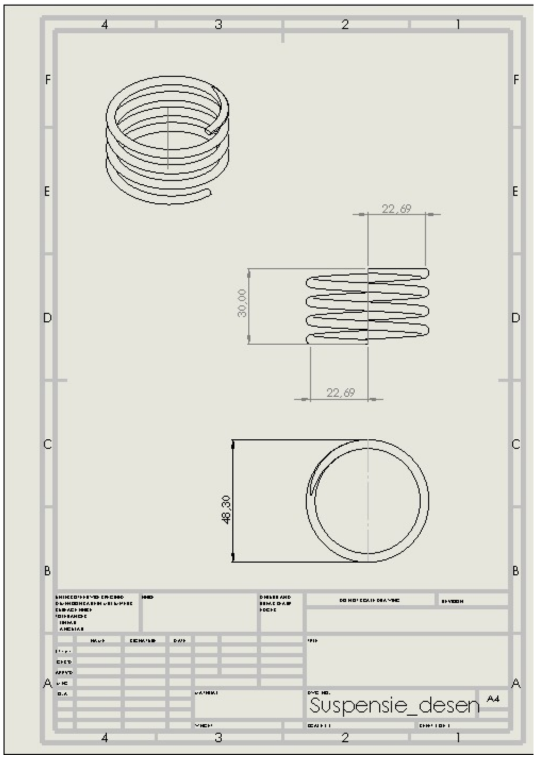

Figure 3: 1.03

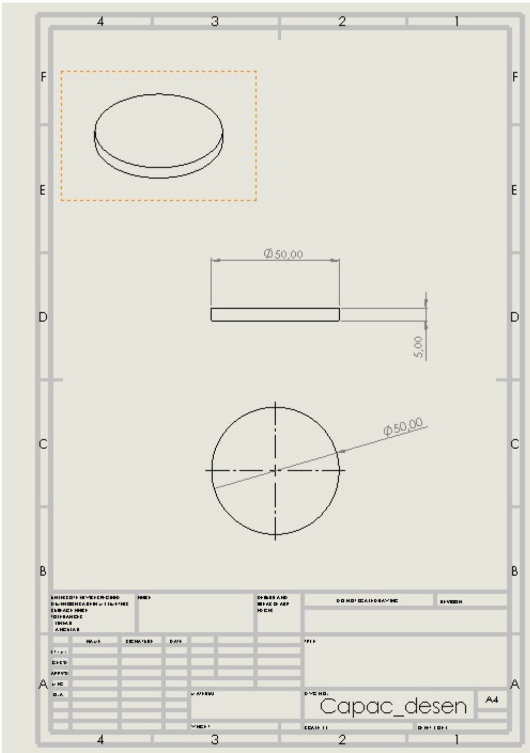

Figure 4: 1.04

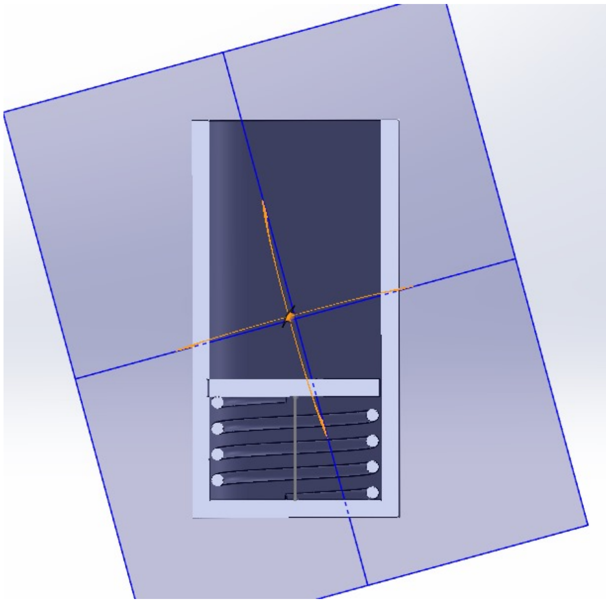

Figure 5: 1.05

5. Explanation: On the sub-probes, the main board acts as the central controller for the CanSat, processing sensor data and controlling the transmitter. The temperature/pressure sensor measures the temperature and pressure of the immediate environment during descent, the pollution sensor compiles data on the quantities of ozone and CO2 gasses present, and the transmitter sends the respective data back to the ground station. The battery provides power to the CanSat during the flight, and the sub probes are stored insomuch as they fit into each other, creating an efficient compact design during the storage process - once deployed, the natural aerodynamic properties of each individual sub probe will allow them to scatter randomly, creating a wide spread for information gathering. The principal CanSat housing unit consists only a small spring and a small holding pin that mechanically deploys the sub-probes once the main body has exited the rocket tube, dispersing the sub-probes almost immediately after initial deployment of the principal CanSat. Each of the three sub probes are comprised of a central component, which houses the main electrical circuits and sensors, and an offset wing that transposes the gravitational acceleration into angular momentum, slowing and dispersing the movement of each individual sub-probe. The principal CanSat is detailed in addendi 1.02-1.05, while the sub-probes are filed under Fig 1.01.

2.3. ELECTRICAL DESIGN

1. Electrical Interface: An electronic drawing is provided in Fig 1.06, detailing the exact positioning of the respective components within the CanSat. The exact electrical components and their functions are as follows: the main circuit board is the Arduino mini, measuring at 17x33mm, the transmitter and receiver is an HC-12 measuring at 27.8x14.4mm, the battery is an unmarked 5v Li-On battery with a 1000mAh capacity measuring at 15x15mm, the gas pollution sensor is a MQ135, measuring at 20x32 mm, and the temperature/pressure sensor is a BMP280, measuring at 12.50x16.50mm. Due to the size and individual power constraints of the 3 sub probes, we have decided to forgo the GPS recovery module, as it is optional; recovery options will be elaborated in chapter 2.6. A detailed diagram of the electrical interface can be found in Fig 1.6

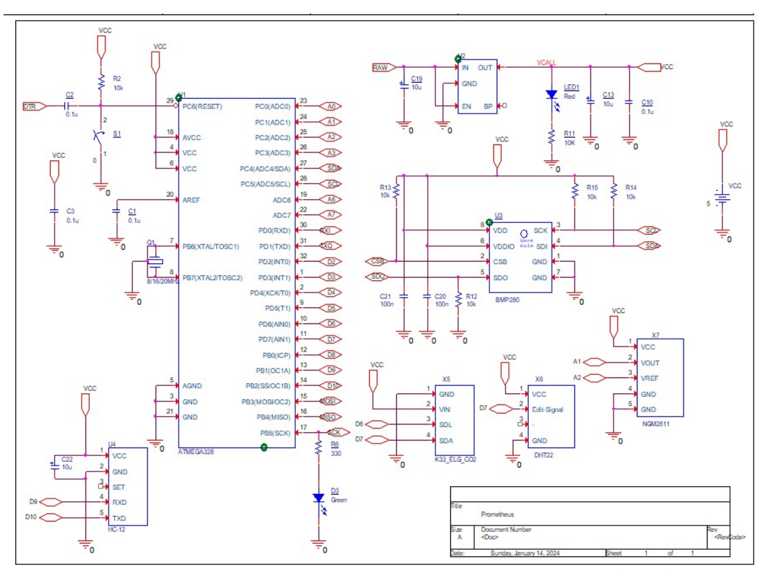

Figure 6: 1.06

2. RF Link: The CanSat uses an HC-12 RF transmitter for data communication with the ground station at distances of up to 1100 meters. The rate of transmission of the downlink is 5000 bps, with identical uplink speeds, and the transmission range is 433.4-473.0MHz, with data packets being sent once every 5 seconds in order to keep power draw to a minimum.

2.4. POWER BUDGET

- Arduino Mini: 4.5 mA
- HC-12 Transmitter: 200mA peak consumption, 80mA average consumption.
- MQ135 Gas Sensor: 150mA peak consumption, 100mA average consumption
- BMP280 Temperature and Pressure sensor: 2.74 µA average consumption
- Battery: 1.1 W

4. Power Consumption and Duration: The total power consumption of each subprobe of the CanSat is estimated to be 182.5 mA, while each individual battery has a capacity of 2000mAh, which should provide enough power for the CanSat to operate for a minimum of 4 hours, accounting for peak consumptions of the MQ135 and HC-12 modules.
5. Battery: Each individual sub-probe uses one (1) 2000mAh Li-Ion battery.

2.5. SOFTWARE DESIGN

1. Software Design: The CanSat's software is designed to control the various electronic components and manage the data collected by the sensors. The main board micro controller runs the software and controls the sensors, transmitter, and other components. The CanSat's software collects data gathered in real time from the aforementioned sensors specified in section 2.3 and transmits them every 5 seconds in regular intervals.
2. Data Gathering and Storage: The CanSat is expected to gather approximately 500 KB of data per flight distributed across the three probes. The data is stored on-board the CanSat unit through the Arduino Pro Mini unit, subsequently transmitted to the ground control unit through the HC-12 radio module.
3. Programming Language and Development Environment: The CanSat software is written primarily for the Arduino platform in Python with small segments written in C, and is developed using the Atom IDE development environment.

2.6. RECOVERY SYSTEM

1. Description: The CanSat sub-probes are designed to deploy and descend in a slow and controlled manner without the use of a specifically designed recovery system. Due to their unique nature, the wing-like structure affixed to the center of gravity that houses the main electronics acts as a slow descent mechanism in of itself, detailed in both sections 2.2 and Fig 1.03. The main CanSat probe body is designed to be discarded, the housing, spring and pin being printed entirely out of biodegradable PLA plastic - due to our unique constraints, we found that a recovery system would add both weight and complexity to the system, though a parachute for easier recovery is not entirely out of the question - however we do not have any significant plans for its integration at the current time. Recovery of the subprobes without an integrated GPS system certainly complicates the recovery process, but through use of transmitter hunting techniques, we can locate each individual probe through the intensity of their respective signal strengths, thereby foregoing the need for an additional component that adds both weight and electrical strain to our project.

2. Picture: A picture of the recovery system design is provided in addendi 1.01-1.05 below.
3. Expected Flight Time: The CanSat's estimated flight time is 4 minutes, including the time required to attain maximum altitude and the subsequent descent back to earth, though this will vary entirely on the exact deployment method and launch altitude, as we have verified with the coordinators that it may vary.

2.7. GROUND SUPPORT EQUIPMENT

1. Equipment: The ground support equipment for the CanSat mission consists of a computer, an antenna, and a transceiver for communication with each of the 3 sub-probes included in the design. The computer receives and compiles the data received into a three-dimensional or two-dimensional graph properly expressing the information received into a visual format for easier human processing.
2. Software Design: The ground software is responsible for the aforementioned graphing system, collecting the data relayed by the sub-probes and displaying it in a visually pleasant manner.
3. Data Handling: The received data is stored in a standard .txt file within the computer, and afterwards can be viewed through a dedicated program for easier viewing.
4. Transmitter Frequency: The transmitter frequency that is present within all CanSat sub-probes is between 433.4-473.0MHz. Through EU standard regulation, the aforementioned frequency range is free use - therefore, we will assign a specific frequency to each probe, presently assigned as 434.0, 435.0, and 436.0 mHz respectively, however we intend to remain flexible as unexpected radio interference is always a very common likelyhood on the day of the launch.

3 PROJECT PLANNING

3.1. Time schedule of the CanSat preparation

Our priorities for the CANSAT development program are as follows, with a more in depth detail found in section 3.4

- Preliminary Design Review (15-21 January)
- Acquisition of Fabrication Components (21-25 January)
- Initial Fabrication and Testing (25 January - 15 February)
- Electronic Assembly and Testing (15 February - 22 February)
- Final Test Phase (22 February - Launch Date (March, TBD))

Schedule is subject to change, and is regularly updated to properly reflect any and all changes to schedule modification.

3.2. RESOURCE ESTIMATION

3.2.1. BUDGET

Below is an approximate estimation of the required resources

Component	Cost (€)
Structure (Metallic parts and structural shields)	30
Transmitter board	15
Controller board	50
Sensor board (Including all the provided sensors and electronic components)	30
Additional materials	60
Additional equipment	40
Launch site fees	50
Total budget	**295**

The above budget is subject to change, as any number of technical mishaps may add to the total spending budget (detailed further in subsection 3.5).

3.2.2. EXTERNAL SUPPORT

The CanSat mission receives support from the Babes-Bolyoi and UTCN universities through the Babes-Bolyoi assistance via use of their 3d printer for project development and processing, and extensive technical assistance and expertise from the Professor body of the two respective universities.

The possibility of sponsorship is still an open topic, so the above list is subject to change. We may choose open a public crowdfunding option on our website and Instagram through sites such as Patreon, though currently there are no immediate plans to do so.

3.3. TEST PLAN

In our primary testing phase, we intend to test the product through a series of timed drops, in varied weather conditions, utilising the natural height of our universities' respective buildings in order to fine-tune our initial descent rate eventually moving on to higher natural elements that exceed said buildings' height (such as cliff sides), thus achieving our desired rate of between 6 and 12 meters per second. During this phase, we intend to use ”dummy” sub probes, mass printed prototypes with small variations and dummy weights to simulate the electronic components of our final design. After proper descent parameters have been confirmed, we will test deployment of the housing cylinder with 3 sub probes contained within. After shock tests and blunt force tests have been concluded, we intend to execute a series of preliminary basic data transmission, aquisition, and processing testing, consisting of the assembly of the HC-12 on the Arduino Mini, along with all sensors to verify functionality to minimize risk of failure. Finally, we will test the shock integrity of the electronic components by assembling a prototype with the final version of the electronic components, and subjecting it to the same tests in our primary testing phases, observing descent from a height of 50 to 60 meters from natural elements. The possibility of a rocket deployment as an initial test exists, as we have procured a launch vehicle - however, due to a lack of authorization as of the writing of this article, we may be forced to abandon this line of testing.

3.4. TIME MANAGEMENT

The general time frame that we are currently working with is between the dates of the 14th of January until an undisclosed date in May 2024, as per the current CANSAT RCRC guidelines. With this information available to us, we intend to structure our development around the timeline detailed in subsection 3.1, with small margins of error granted for eventual mishaps or interruptions. Firstly, acquiring our source of 3D printed prototypes of our descent vehicles, testing the descent rate and modifying the design to comply with the current RCRC guideline, between 6 and 12 meters per second. We estimate this phase of prototyping will take approximately 2 weeks, though we will allow a week extra as a margin of error. In the meantime, we will await our orders for our specified parts, which we estimate to take approximately 3-4 weeks - once our electronic components arrive, we can begin testing the integrated product, which will take another approximate 1-2 weeks. Once electrical systems and integration has been completed and we have a fully working prototype, we will attempt to iron out or improve our design with whatever remaining time we have, though we fully expect problems and kinks to appear, leaving the prospect of any surplus time one we find dubious at best. Time management and framework in a digital environment in our team is viewed as largely unnecessary, as we manage to physically document our progress in a series of journals, as well as maintain bi-weekly meetings, a core value we maintained throughout the entirety of the CANSAT program thus far.

3.5. RISK MANAGEMENT

- Risk 1: Delays in obtaining materials
  Mitigation: We hope to immediately order the required electronics components through a series of online marketplaces, as those are the only materials that may be affected by this risk, as the physical component will almost entirely be constructed through 3D printing. If our orders online are delayed, we have a backup physical location with the proper items in stock, though the aforementioned location has increased prices, leaving it as only a last resort option.
- Risk 2: Unforeseen technical difficulties or physical faults
  Mitigation: On the electronics side of our project, as mentioned in Risk 1, we have a physical vendor that will allow us to replace any broken or faulty components on a short notice, granted at an additional cost. On the digital side of our project, we plan to develop at least one alternative to the code that will be implemented during testing and on launch day. On the physical component side of our project, we hope to manufacture spare parts in order to assure our prototype is in perfect working condition on launch day, and test them sufficiently insomuch we minimize the event of component failure. In the event that frequency overlap occurs between our sub probes or other devices on launch day, we maintain the capacity to shift the operating frequency with little to no effort of each sub probe, thus greatly reducing the possibility of loosing connection to our sub probes.
- Risk 3: Weather delays on launch day
  Mitigation: As weather delays can be seen ahead of time through modern meteorological methods, the eventual rescheduling of our flight plan will thankfully pose the least threatening risk of our three risks, though the eventual planning of at least one backup date on the event that weather forecasting is not accurate is entirely within our current plan.

4 DATA ANALYSIS AND OUTREACH

4.1. DATA ANALYSIS PLAN

The data analysis plan for the CanSat mission is designed to process and interpret the data collected by the various sensors on board the CanSat. The following software and tools can be used for data analysis:

- Data acquisition software, data processing software and statistical analysis software: We

intend to use https://www.labcenter.com/iotbuilder/ for direct control and communication through the HC-12 radio control unit.
- Data visualisation software: Data visualisation is slated to be produced by our own team, simulated through blender to create a 3d experience that is easy to read and understandable. -
- GIS software: Geographic Information System software will be charted through a custom software using google maps plugins.

These separate elements combined will render a 3 dimensional map of temperature, gas, and pressure analysis, creating the opportunity to glean not only an altitude based interpretation of the main project parameter - air pollution and temperature dynamics - but also a spatial interpretation, giving us the possibility to tell where said pollutants may be concentrated. While we have foregone the use of a GPS system, the HC-12 module allows for inter-probe communication, and based on time to ping between probes, as well as time to ping for each individual probe to ground, we can retain a good idea as to the position of each CANSAT sub probe. The data analysis plan is still subject to change, as software implementation research is still underway.

4.2. OUTREACH PROGRAM

- Presentations: Our team intends to present the aforementioned project through a series of community demonstrations within the confines of the UTCN and UBB universities, through the assistance of our resident professors. We intend to demonstrate that the technology and software we use is reliable, safe, efficient, and can fulfill the projected goals, providing an educational experience and a solid tangible proof of concept for both individuals seeking general development within the scientific industry as well as possible investors and donors.
- Social Media: We currently have an active and engaged following through Instagram and Whatsapp social media groups, with regular stories and semi-regular posts, detailing our activities as a team, and a few of our future plans through "teasers", thus keeping our community updated and informed as to our future and current developments. In the near future, we plan to set up linked Facebook and LinkedIn profiles for our host startup, RedShift Aerospace Industries, along with auxiliary outreach programs through popular platforms such as Discord and TikTok. Our Instagram page can be found at https://www.instagram.com/redshiftaero/ .
- Websites: Our team has developed a proof of concept website, detailing our team member composition, our design and technical specifications for the CANSAT competition, links to our Instagram and (soon to come) Facebook profiles, and a community outreach email for feedback and communication. Further development of our website in the near future will aim to host video and text based presentations, detailing our activities with a greater degree of precision then can be offered by short-form content that is typically hosted by social media outlets, such as Instagram or Facebook. Our website can be found at https://redshiftareo.com/ .

5 CONCLUSION

5.1. SUMMARY OF THE PDR

In summary, we have formulated a plan to create a CanSat that, when constructed, will develop sub-probes that in turn will allow for an efficient vector analysis of pressure, temperature and pollution diffusion and coefficients respectively.

5.2. RECOMMENDATIONS FOR NEXT STEPS

Our next steps include building the CanSat, further developing our social media, website and demonstration aspects, securing a launch platform, testing the main probe and the respective sub-probes, and developing a software platform that will ensure proper visualization of all data received in real time by the receiver. Additionally, through rigorous testing, we hope to eliminate most if not all major points of failure before the designated launch date.